Heaven's Light is Our Guide



Department of Electrical & Electronic Engineering

Rajshahi University of Engineering & Technology

Final Project

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Course Title : **Biomedical Signals, Instrumentation and Measurements Sessional**

Project Title : EEG-Based Epileptic Seizure Detection

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EEG-Based Epileptic Seizure Detection

Abstract— Electroencephalography (EEG) signal processing is widely used to analyze brain activity and support automated seizure detection. This study presents a biomedical signal processing framework for EEG analysis using the University of Bonn EEG dataset. Following the assignment protocol, five representative recordings were selected from each of the five EEG classes, resulting in a balanced dataset of 25 single-channel EEG recordings. The framework comprises raw EEG waveform visualization, frequency-domain analysis using Welch's power spectral density (PSD), and extraction of eleven handcrafted features from the time, frequency, and Hjorth domains. Feature distributions, class-wise characteristics, and inter-feature relationships were further investigated using boxplots, heatmaps, and correlation analysis to evaluate their discriminative capability before classification. A Random Forest classifier was subsequently employed to perform three binary classification tasks (A vs. E, A vs. D, and C vs. E) using an 80% training and 20% testing split. The framework achieved 100% accuracy, precision, recall, and F1-score for all evaluated tasks. Feature importance analysis further demonstrated that spectral band-power features and Hjorth parameters contributed most significantly to seizure discrimination. These findings highlight the effectiveness of integrating conventional EEG signal processing techniques with machine learning for automated seizure detection.

I. Introduction

Biomedical signal processing is a fundamental area of biomedical engineering that focuses on analyzing physiological signals to extract meaningful information for diagnosis, monitoring, and decision support. Among various physiological signals, electroencephalography (EEG) has received significant attention because it provides a non-invasive measurement of the brain's electrical activity with high temporal resolution. Owing to its ability to capture dynamic neural behavior, EEG has become an important signal for studying brain function and developing computer-aided neurological diagnostic systems.

Raw EEG recordings are highly complex, non-stationary, and often contaminated by various sources of noise and artifacts. Consequently, meaningful interpretation of EEG signals requires appropriate signal processing techniques. Time-domain analysis provides insight into the statistical characteristics of the recorded signals, while frequency-domain analysis reveals the distribution of spectral energy across different frequency components. Furthermore, feature extraction transforms high-dimensional EEG recordings into compact and informative representations that preserve discriminative characteristics while reducing computational complexity. These signal processing techniques provide the foundation for subsequent automated analysis and classification.

Recent advances in machine learning have further enhanced EEG analysis by enabling automatic recognition of complex patterns that are difficult to identify through manual inspection. When informative signal processing features are combined with supervised learning algorithms, reliable classification systems can be developed for distinguishing different physiological and pathological

brain activities. Such integrated approaches have become increasingly important in computer-aided seizure detection and neurological signal analysis.

The objective of this study is to investigate a biomedical signal processing framework for EEG-based seizure detection by integrating signal analysis and machine learning techniques. The study demonstrates how waveform visualization, spectral analysis, feature extraction, and supervised classification can be combined to analyze EEG signals and distinguish different brain activity patterns, highlighting the importance of signal processing in modern biomedical applications.

II. Materials and Methodology

A. Dataset Description:

The University of Bonn EEG dataset [1] is a publicly available benchmark dataset widely used for biomedical signal processing and epileptic seizure detection research. It comprises 500 single-channel EEG recordings equally distributed among five classes (A-E), with 100 recordings per class. Each recording contains 4,097 samples acquired at a sampling frequency of 173.6 Hz. The five classes represent different physiological conditions, ranging from healthy brain activity to active epileptic seizures, as summarized in Table 1. For this project, five recordings from each class were selected, resulting in a balanced dataset of 25 EEG recordings.

Table 1: Description of the Bonn University EEG dataset.

Class	Subject	Condition	Description
A	Healthy volunteers	Eyes open	Baseline EEG activity
B	Healthy volunteers	Eyes closed	Normal EEG with stronger alpha rhythm
C	Epileptic patients	Inter-ictal	Seizure-free activity (opposite hemisphere)
D	Epileptic patients	Inter-ictal	Seizure-free activity (epileptogenic zone)
E	Epileptic patients	Ictal	Active epileptic seizure activity

B. Signal Processing:

B.1 Raw EEG Signal Visualization:

Raw EEG waveform visualization was performed to examine the temporal characteristics of the acquired EEG signals prior to further analysis. Five representative recordings from each EEG class were visualized in the time domain using the original sampling frequency of 173.6 Hz. Figures 1-5 present the representative raw EEG waveforms for Classes A-E, respectively. This step preserves the original signal characteristics and establishes the input for the subsequent signal processing procedures.

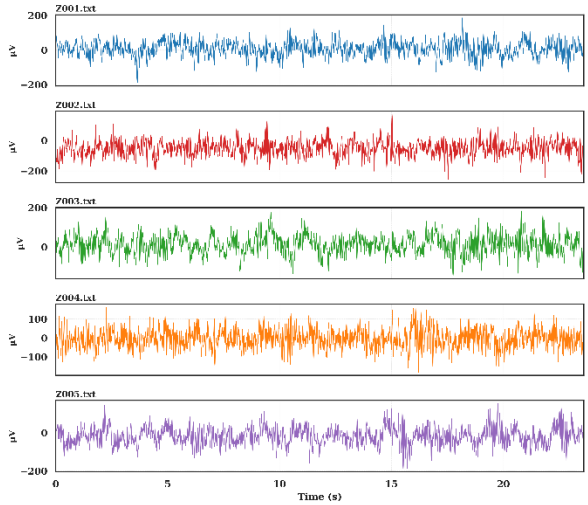


Figure 1. Raw EEG waveforms of Class A

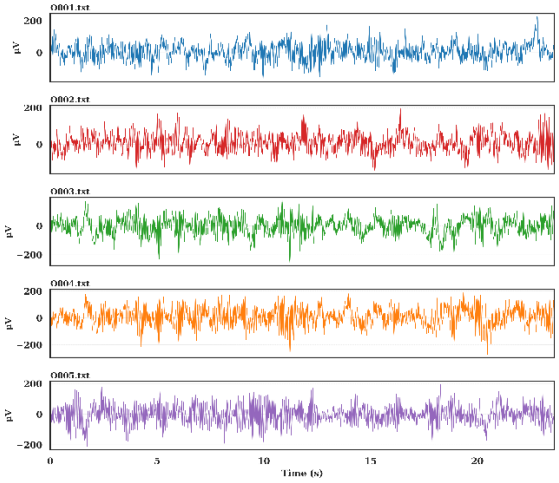


Figure 2. Raw EEG waveforms of Class B

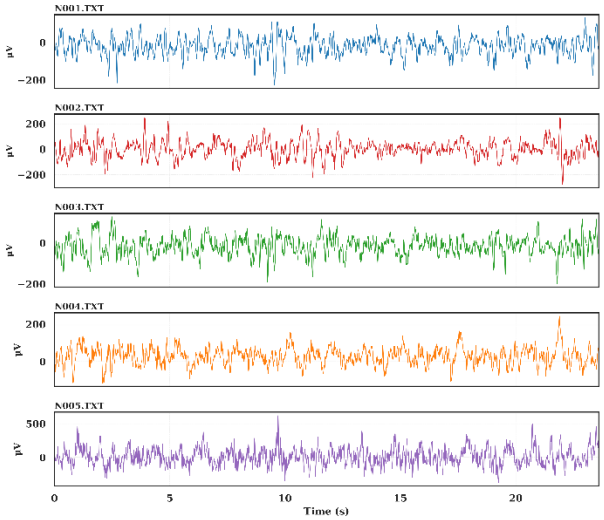


Figure 3. Raw EEG waveforms of Class C

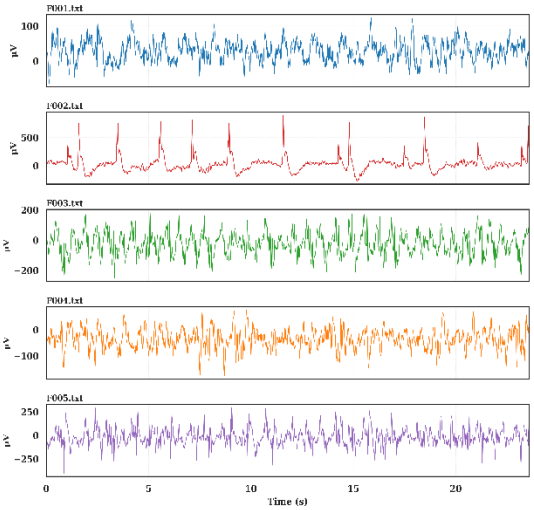


Figure 4. Raw EEG waveforms of Class D

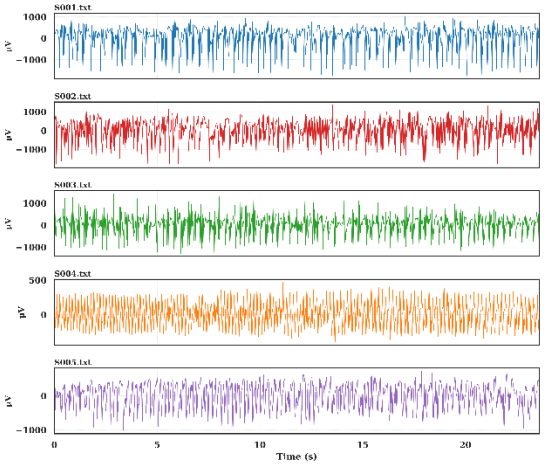


Figure 5. Raw EEG waveforms of Class E

B.2 Welch Power Spectral Density Estimation

The frequency-domain characteristics of the EEG signals were analyzed using the Welch power spectral density (PSD) estimator [2]. Welch's method estimates the signal power by dividing each EEG recording into overlapping segments, applying a Hann window to each segment, computing the Fourier transform, and averaging the resulting periodograms. This approach provides a smoother and more reliable spectral estimate than a conventional periodogram while reducing estimation variance. For an EEG signal x(n), the PSD estimated by Welch's method is given by

$$P_{Welch}(f) = \frac{1}{K}\sum_{k=1}^{K} P_k(f)$$

where $P_k(f)$ is the periodogram of the k^{th}overlapping segment and K is the total number of segments. In this project, a Hann window, 512-sample segment length, and 50% overlap were used to estimate the PSD of each EEG recording. The PSDs of the five recordings belonging to the same class were then averaged to obtain the representative spectral profile of that class. The resulting mean PSD curves for Classes A-E are presented in Figure 6.

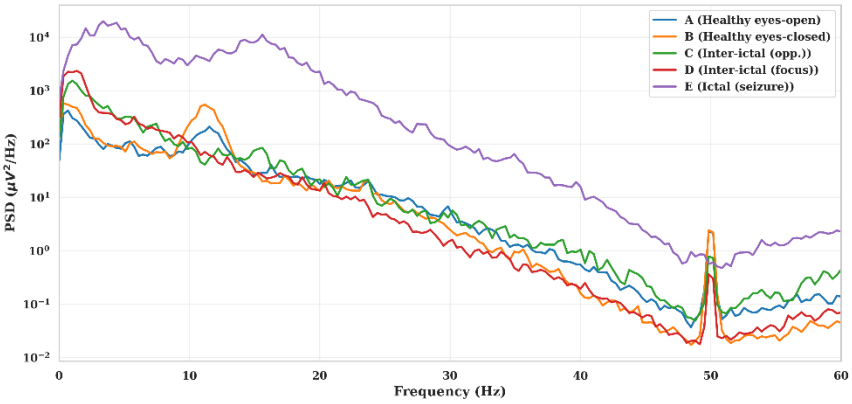


Figure 6. Mean power spectral density of the five EEG classes estimated using Welch's method.

B.3 Feature Extraction

To characterize the EEG recordings, a total of 11 hand-crafted features were extracted from each signal. These features capture complementary information from the time, frequency, and Hjorth domains. Specifically, four statistical time-domain features (mean, standard deviation, kurtosis, and skewness), five spectral band-power features (delta, theta, alpha, beta, and gamma), and two Hjorth parameters (mobility and complexity) were computed for every EEG recording. The mean and standard deviation describe the central tendency and variability of the EEG signal, while skewness and kurtosis quantify the asymmetry and peakedness of the amplitude distribution, respectively. The mean and standard deviation are defined as

$$\mu = \frac{1}{N}\sum_{n=1}^{N} x(n), \qquad \sigma = \sqrt{\frac{1}{N}\sum_{n=1}^{N}(x(n)-\mu)^2}$$

Frequency-domain features were extracted from the Welch PSD. The power within each EEG frequency band was computed by integrating the PSD over the corresponding frequency range,

$$P_{band} = \int_{f_1}^{f_2} P_{Welch}(f)\, df$$

and subsequently converted to logarithmic scale (dB). In addition, the Hjorth parameters were calculated to characterize the temporal dynamics of the EEG signals,

$$Mobility = \sqrt{\frac{Var(\dot{x})}{Var(x)}}, \quad Complexity = \frac{Mobility(\dot{x})}{Mobility(x)}$$

B.4 Feature Visualization and Analysis

To comprehensively evaluate the discriminative characteristics of the extracted EEG features, three complementary visualization techniques were employed. These visualizations provide qualitative insight into the statistical behavior, class-specific distributions, and inter-feature relationships prior to classifier training. Figure 7 presents boxplots of the eleven extracted features for all five EEG classes. Boxplots summarize the median, interquartile range, and overall data dispersion while highlighting potential outliers..

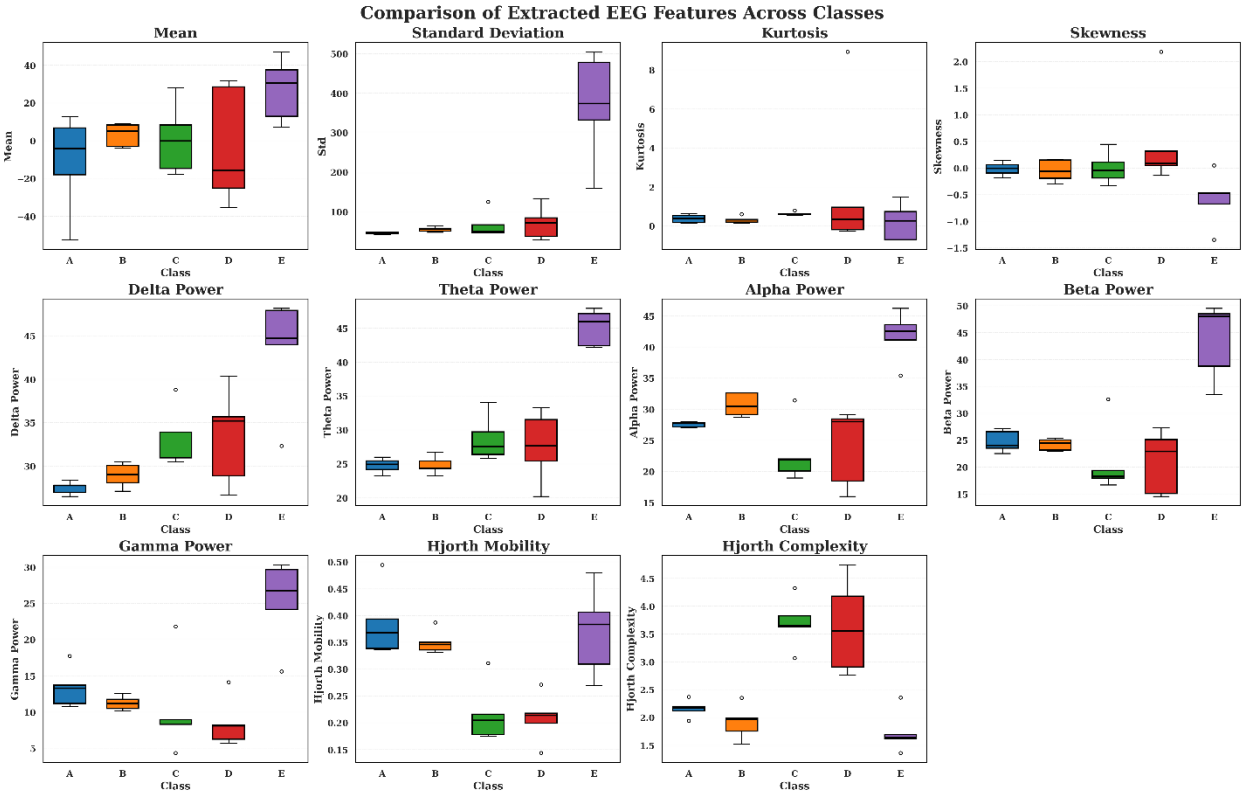


Figure 7. Boxplots showing the distributions of the eleven extracted EEG features across the five EEG classes.

To provide an overall comparison of feature characteristics among the EEG classes, the class-wise mean feature heatmap is presented in Figure 8. The heatmap illustrates the average value of each extracted feature for every class using color intensity, allowing rapid identification of class-dependent variations and dominant feature patterns. This representation offers a concise summary of the overall feature distribution and assists in identifying features that exhibit distinct responses across different physiological conditions.

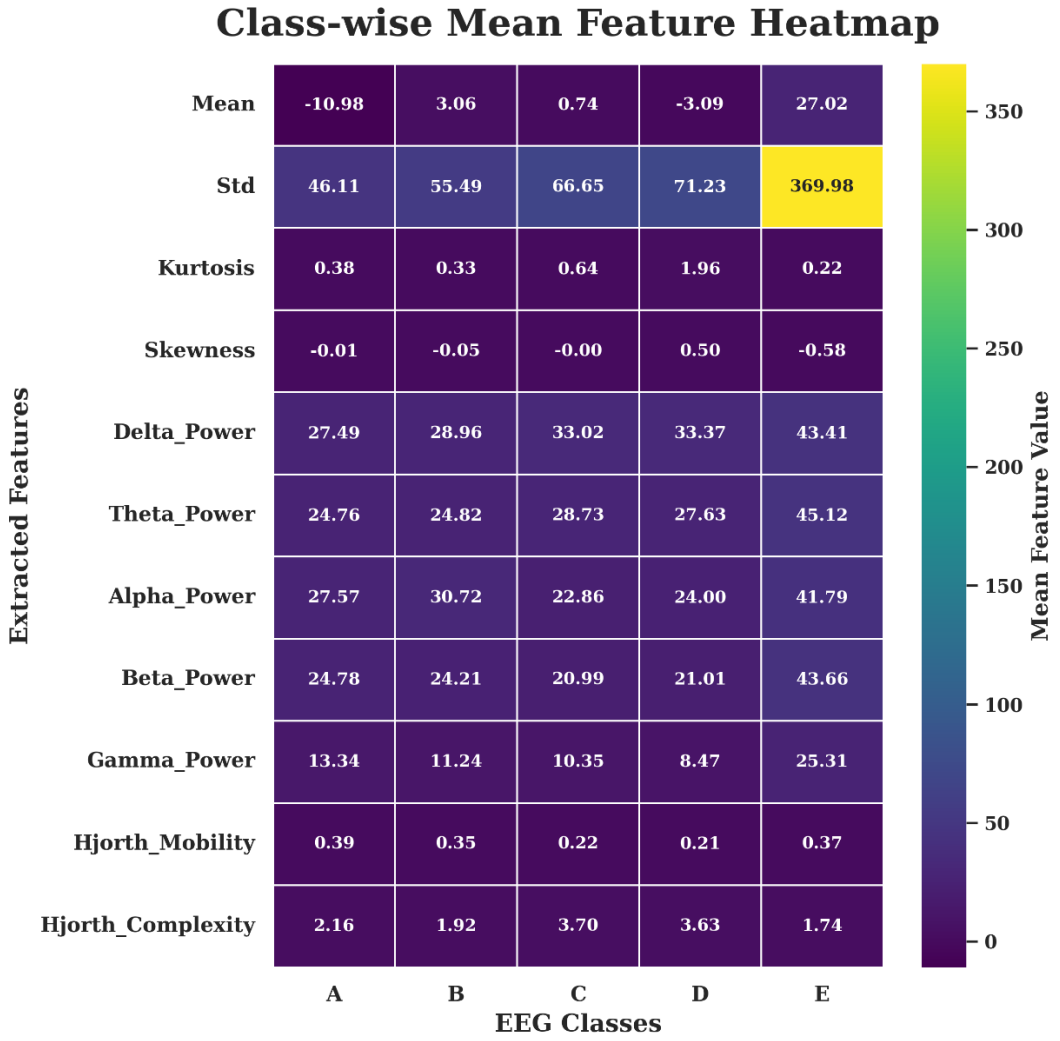


Figure 8. Class-wise mean values of the extracted EEG features. Each cell represents the average value of a feature computed from all recordings belonging to a specific EEG class.

Finally, Figure 9 presents the Pearson correlation matrix of the extracted features. The correlation analysis quantifies the linear relationship between feature pairs and provides insight into feature dependency and redundancy. Highly correlated features may contain overlapping information, whereas weakly correlated features are more likely to provide complementary information for classification. Together, these visualization techniques provide a comprehensive understanding of the extracted feature set and establish its suitability for subsequent Random Forest-based EEG classification.

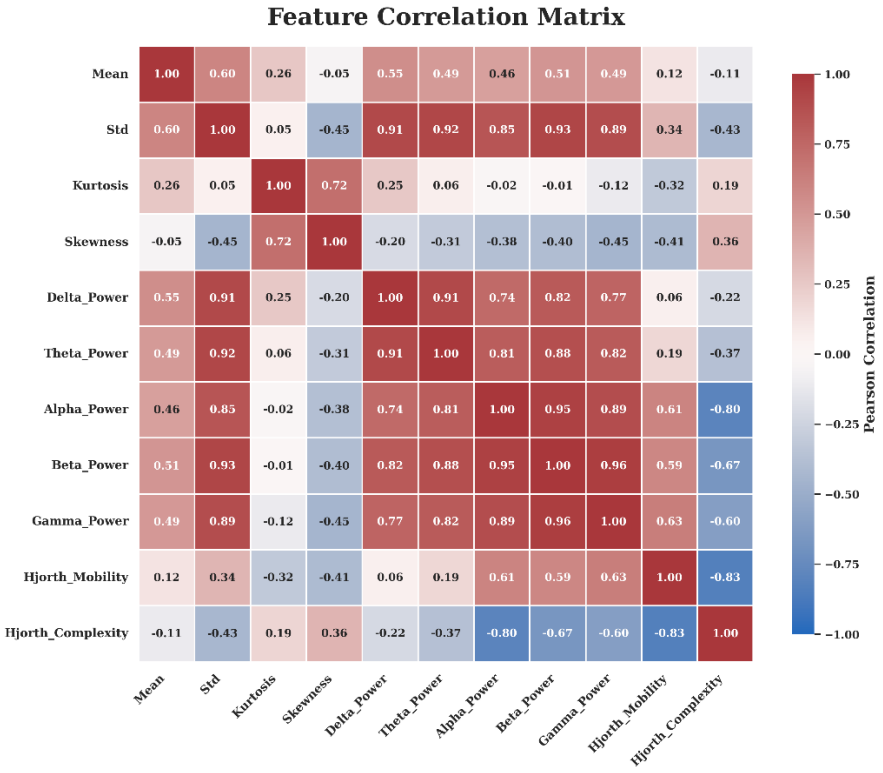


Figure 9. Pearson correlation matrix of the extracted EEG features, illustrating the strength and direction of linear relationships among the extracted features.

B.5 Random Forest Model Training

The extracted feature matrix was used to train a Random Forest (RF) classifier [3] for EEG signal classification. Random Forest is an ensemble learning algorithm that constructs multiple decision trees using randomly selected subsets of the training data and features. During inference, each decision tree independently predicts the class label, and the final classification is determined by majority voting among all trees. This ensemble strategy improves classification robustness by reducing overfitting and enhancing generalization.

Let $T_i(x)$ denote the class prediction of the ith decision tree for an input feature vector x. The final prediction of the Random Forest classifier is obtained as

$$\hat{y} = mode\{T_1(x), T_2(x), \dots, T_N(x)\}$$

where N represents the total number of decision trees.

Following the assignment specification, the 25 EEG recordings were divided into 80% training (20 recordings) and 20% testing (5 recordings) using a stratified random split to preserve the class distribution. Only the training set was used to construct the classifier, while the test set remained completely unseen during model development and was reserved exclusively for performance evaluation. The Random Forest classifier employed the Gini impurity criterion for node splitting and consisted of 100 decision trees.

B.6 Performance Evaluation

The performance of the trained Random Forest classifier was evaluated using the independent test set, which was not used during model training. In accordance with the project specification, three binary classification tasks were considered: Class A vs. Class E, Class A vs. Class D, and Class C vs. Class E. These classification scenarios assess the capability of the classifier to distinguish between healthy, inter-ictal, and ictal EEG signals. Classification performance was assessed using four widely adopted evaluation metrics: accuracy, precision, recall, and F1-score. The evaluation metrics are defined as

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}}$$

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}$$

$$\text{F1-Score} = \frac{2\text{TP}}{2\text{TP} + \text{FP} + \text{FN}}$$

where TP, TN, FP, and FN denote the numbers of true positives, true negatives, false positives, and false negatives, respectively. These metrics provide a comprehensive assessment of the classifier and were used to evaluate the three binary EEG classification tasks.

III. Results and Discussion

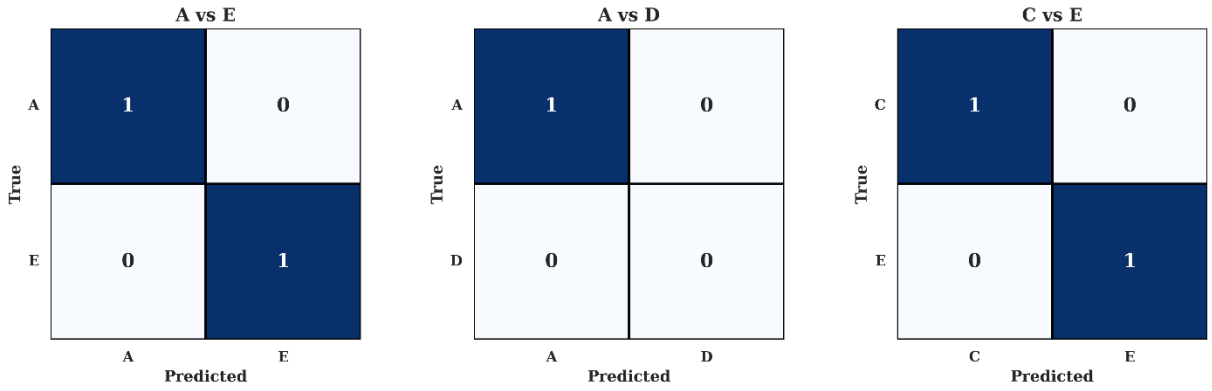
A. Classification Performance

The classification performance of the Random Forest classifier was evaluated using the independent test set for three binary classification tasks: Class A vs. Class E, Class A vs. Class D, and Class C vs. Class E. Classification performance was assessed using accuracy, precision, recall, and F1-score, with the corresponding results summarized in Table 2.

Table 2: Performance of the Random Forest classifier for the binary EEG classification tasks.

Classification Task	**Accuracy (%)**	**Precision (%)**	**Recall (%)**	**F1-score (%)**
A vs. E	100.00	100.00	100.00	100.00
A vs. D	100.00	100.00	100.00	100.00
C vs. E	100.00	100.00	100.00	100.00

As presented in Table 2, the classifier achieved 100% for all evaluation metrics across the three binary classification tasks. These findings indicate that the extracted statistical features, spectral band-power features, and Hjorth parameters effectively capture the discriminative characteristics of the EEG signals, enabling accurate separation of healthy, inter-ictal, and ictal brain activities. The high classification performance is consistent with the distinct feature distributions observed during the feature analysis stage.



(a) (b) (c)

Figure 10. Confusion matrices of the Random Forest classifier for the binary EEG classification tasks: (a) Class A vs. Class E, (b) Class A vs. Class D, and (c) Class C vs. Class E.

The corresponding confusion matrices are shown in Figure 10. No false-positive or false-negative predictions were observed for any of the evaluated classification tasks, indicating complete agreement between the predicted and true class labels for the available test samples. In particular, the A vs. E and C vs. E tasks achieved perfect discrimination between healthy/inter-ictal and ictal EEG recordings, while the A vs. D task also correctly classified the available test sample.

B. Feature Importance Analysis

Feature importance analysis was performed to identify the contribution of each extracted EEG feature to the Random Forest classifier. This analysis complements the feature visualization presented in Section II.B.4 by providing a model-driven interpretation of the extracted descriptors. While the boxplots, class-wise mean heatmap, and correlation matrix illustrated feature distributions and inter-feature relationships, feature importance quantifies their relative influence on the classification decisions.

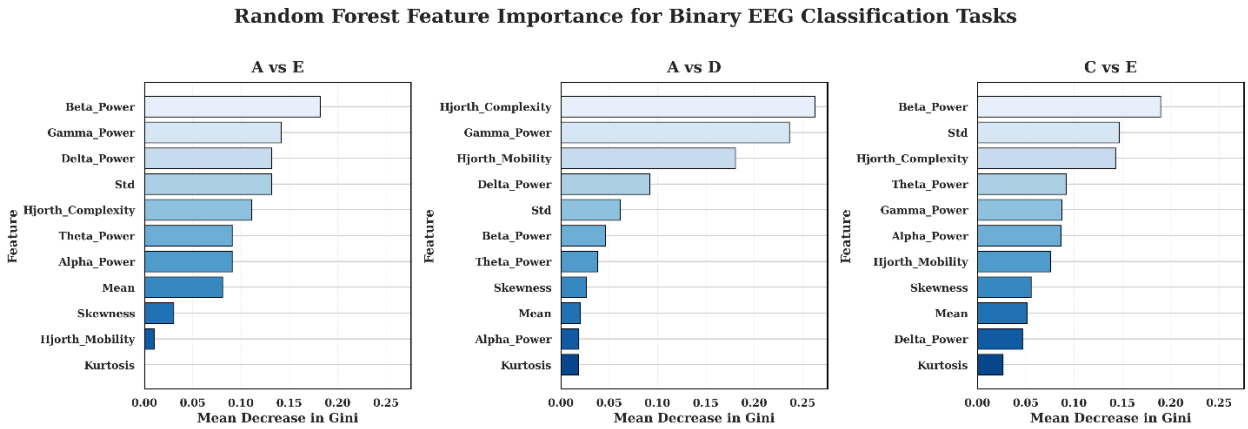


Figure 11. Random Forest feature importance for the three binary EEG classification tasks

As shown in Figure 11, the frequency-domain features consistently exhibited the highest importance across the three binary classification tasks. For A vs. E and C vs. E, beta_power was

the most influential feature, followed by gamma_power, Std, and other spectral band-power features. In contrast, Hjorth_Complexity, gamma_power, and Hjorth_Mobility contributed most strongly to the A vs. D classification. Conversely, the first-order statistical descriptors, including mean, kurtosis, and skewness, showed relatively low importance in all tasks. These findings are consistent with the observations in Section B.4, where spectral and Hjorth-domain features exhibited greater class separability than the basic statistical descriptors.

Overall, the feature importance analysis confirms that the extracted spectral and temporal dynamic features provide the primary discriminative information for binary EEG classification and supports the excellent classification performance reported in Section III.A.

C. Discussion

The obtained results demonstrate that the signal processing framework effectively distinguishes healthy, inter-ictal, and ictal EEG signals using a combination of handcrafted features and a Random Forest classifier. The extracted statistical, spectral, and Hjorth features provide complementary information, while the feature visualization and importance analyses consistently indicate that frequency-domain descriptors and Hjorth parameters contribute most significantly to the classification process. These findings highlight the importance of combining multiple signal-processing domains for reliable EEG characterization. Although perfect classification performance was achieved for all three binary classification tasks, the evaluation was conducted on a limited subset of the Bonn EEG dataset following the assignment specification. Therefore, the reported results should be interpreted within the scope of the experimental protocol rather than as evidence of generalized clinical performance

IV. Conclusion

This project presented a biomedical signal processing framework for EEG-based seizure detection using the University of Bonn EEG dataset. The workflow integrated raw EEG visualization, Welch power spectral density analysis, handcrafted feature extraction, feature analysis, and Random Forest classification. The extracted statistical, spectral, and Hjorth features effectively characterized the EEG signals, enabling the Random Forest classifier to achieve 100% accuracy, precision, recall, and F1-score across the three evaluated binary classification tasks. Although the evaluation was conducted on a limited subset of the dataset, the results demonstrate the effectiveness of combining conventional signal processing techniques with machine learning for automated EEG signal classification.

References

[1] R. G. Andrzejak, K. Lehnertz, F. Mormann, C. Rieke, P. David, and C. E. Elger, "Indications of nonlinear deterministic and finite-dimensional structures in time series of brain electrical activity: Dependence on recording region and brain state," *Physical Review E,* vol. 64, no. 6, p. 061907, 2001.

[2] P. Welch, "The use of fast Fourier transform for the estimation of power spectra: A method based on time averaging over short, modified periodograms," *IEEE Transactions on audio and electroacoustics,* vol. 15, no. 2, pp. 70-73, 1967.

[3] L. Breiman, "Random forests," *Machine learning,* vol. 45, no. 1, pp. 5-32, 2001.